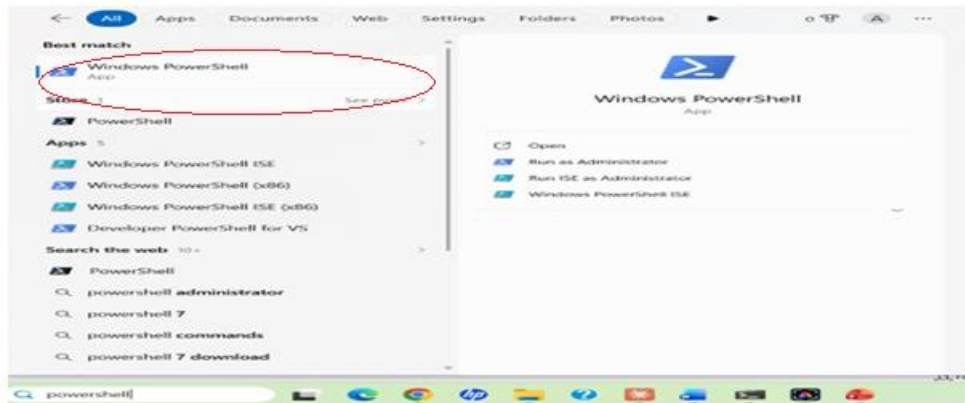
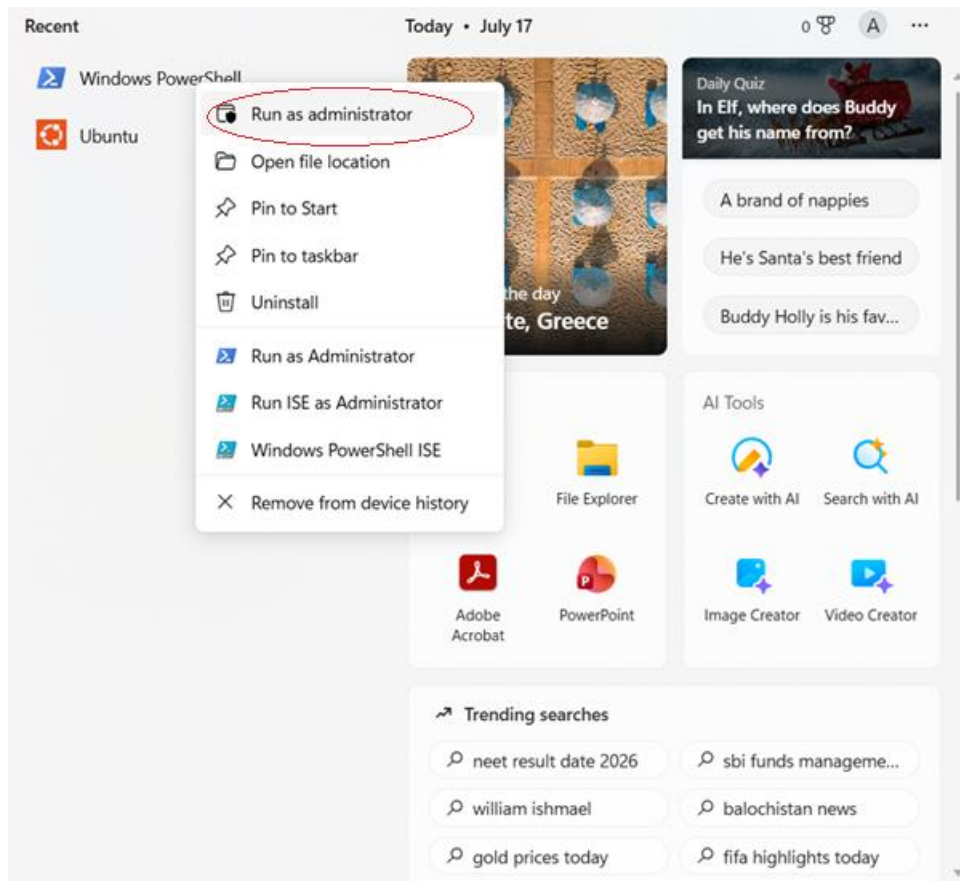


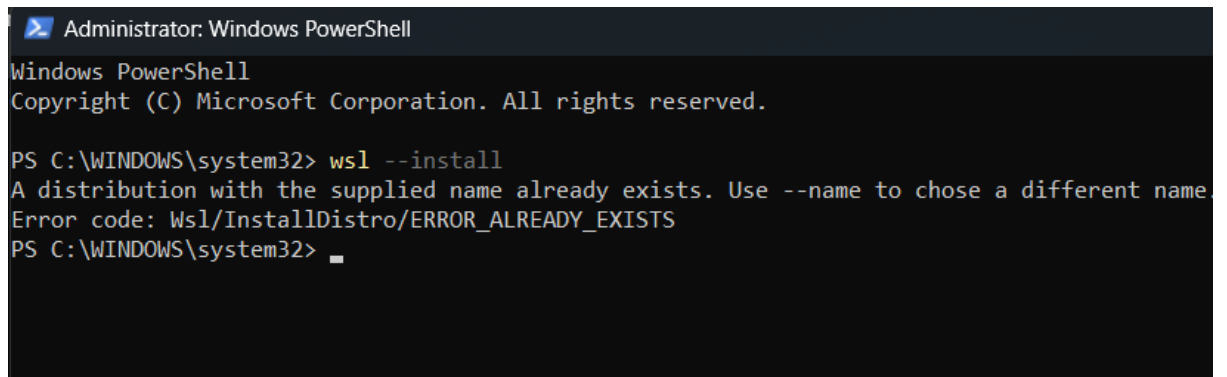
Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.



3. Right-click **Windows PowerShell**.



A screenshot of a Windows PowerShell terminal window titled "Administrator: Windows PowerShell". The window shows the following text: "Windows PowerShell", "Copyright (C) Microsoft Corporation. All rights reserved.", "PS C:\WINDOWS\system32> wsl --install", "A distribution with the supplied name already exists. Use --name to chose a different name.", "Error code: Wsl/InstallDistro/ERROR_ALREADY_EXISTS", and "PS C:\WINDOWS\system32>".

```
Administrator: Windows PowerShell

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> wsl --install
A distribution with the supplied name already exists. Use --name to chose a different name.
Error code: Wsl/InstallDistro/ERROR_ALREADY_EXISTS
PS C:\WINDOWS\system32>
```

Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

Enter new UNIX username: ketana_sindhu

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password: ****

Step 6: Update Ubuntu

```
ketana_sindhu@KetaZone:~$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [417 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [97.2 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 c-n-f Metadata [4616 B]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [159 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [51.2 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3552 B]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [349 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [67.8 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 c-n-f Metadata [396 B]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [9020 B]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [2660 B]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Components [212 B]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 c-n-f Metadata [120 B]
Get:19 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages [1480 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute/main Translation-en [524 kB]
Get:22 http://archive.ubuntu.com/ubuntu resolute/main amd64 Components [395 kB]
Get:23 http://archive.ubuntu.com/ubuntu resolute/main amd64 c-n-f Metadata [32.4 kB]
Get:24 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [16.0 MB]
Get:25 http://archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:26 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4556 kB]
Get:27 http://archive.ubuntu.com/ubuntu resolute/universe amd64 c-n-f Metadata [313 kB]
Get:28 http://archive.ubuntu.com/ubuntu resolute/restricted amd64 Packages [152 kB]
Get:29 http://archive.ubuntu.com/ubuntu resolute/restricted Translation-en [25.8 kB]
Get:30 http://archive.ubuntu.com/ubuntu resolute/restricted amd64 Components [556 B]
Get:31 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages [290 kB]
Get:32 http://archive.ubuntu.com/ubuntu resolute/multiverse Translation-en [127 kB]
Get:33 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Components [50.0 kB]
Get:34 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 c-n-f Metadata [8276 B]
Get:35 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [492 kB]
Get:36 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [118 kB]
Get:37 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:38 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 c-n-f Metadata [5756 B]
Get:39 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [243 kB]
Get:40 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [81.0 kB]
Get:41 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [184 kB]
Get:42 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 c-n-f Metadata [4704 B]
Get:43 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [367 kB]
Get:44 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [70.8 kB]
Get:45 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 c-n-f Metadata [392 B]
Get:46 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.3 kB]
Get:47 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3032 B]
Get:48 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Components [216 B]
Get:49 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 c-n-f Metadata [256 B]
Get:50 http://archive.ubuntu.com/ubuntu resolute-backports/main amd64 Components [212 B]
Get:51 http://archive.ubuntu.com/ubuntu resolute-backports/main amd64 c-n-f Metadata [112 B]
Get:52 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [640 B]
Get:53 http://archive.ubuntu.com/ubuntu resolute-backports/universe Translation-en [300 B]
Get:54 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [216 B]
Get:55 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 c-n-f Metadata [116 B]
Get:56 http://archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 Components [216 B]
Get:57 http://archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 c-n-f Metadata [120 B]
Get:58 http://archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 Components [216 B]
Get:59 http://archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 c-n-f Metadata [120 B]
Fetched 33.7 MB in 6s (5373 kB/s)
123 packages can be upgraded. Run 'apt list --upgradable' to see them.
ketana_sindhu@KetaZone:~$ |
```

Step 7: Install GCC (C Compiler)

```
ketana_sindhu@KetaZone:~$ sudo apt install build-essential
Upgrading:
  build-essential

Installing dependencies:
  libcrypt-dev

Summary:
  Upgrading: 1, Installing: 1, Removing: 0, Not Upgrading: 122
  Download size: 127 kB
  Space needed: 384 kB / 1024 GB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu/resolute/main amd64 libcrypt-dev amd64 1:4.5.1-1 [122 kB]
Get:2 http://archive.ubuntu.com/ubuntu/resolute-updates/main amd64 build-essential amd64 12.12ubuntu2.26.04.2 [5158 B]
Fetched 127 kB in 0s (458 kB/s)
Selecting previously unselected package libcrypt-dev:amd64.
(Reading database ... 42256 files and directories currently installed.)
Preparing to unpack .../libcrypt-dev_1%3a4.5.1-1_amd64.deb ...
Unpacking libcrypt-dev:amd64 (1:4.5.1-1) ...
Preparing to unpack .../build-essential_12.12ubuntu2.26.04.2_amd64.deb ...
Unpacking build-essential (12.12ubuntu2.26.04.2) over (12.12ubuntu2) ...
Setting up libcrypt-dev:amd64 (1:4.5.1-1) ...
Setting up build-essential (12.12ubuntu2.26.04.2) ...
Processing triggers for man-db (2.13.1-1build1) ...
ketana_sindhu@KetaZone:~$
```

Step 8: Verify the Installation

```
ketana_sindhu@KetaZone:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

ketana_sindhu@KetaZone:~$ |
```

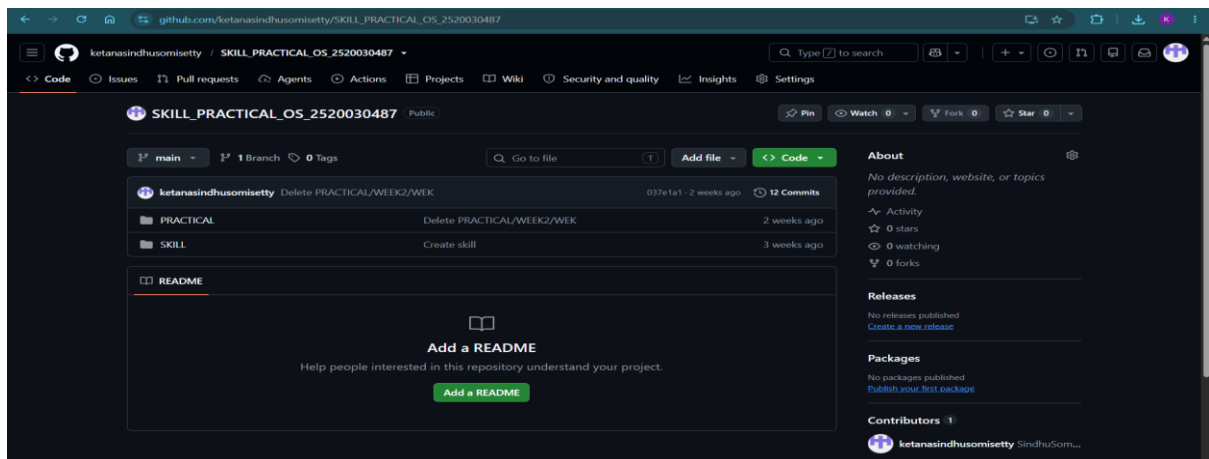
Step 9: Create Your C Program

Example: nano process.c

Step 10: Compile the Program

```
ketana_sindhu@KetaZone:~$ gcc HELLO.c -o hello
ketana_sindhu@KetaZone:~$ ./hello
Hello Linux
ketana_sindhu@KetaZone:~$ |
```

GitHub Repo SS:



Fork program

Code

```
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>

int main()
{
    fork();

    printf("Hello\n");

    return 0;
}
```

```
ketana_sindhu@KetaZone:~$ gcc Fork.c -o fork
ketana_sindhu@KetaZone:~$ ./fork
Hello
Hello
ketana_sindhu@KetaZone:~$ |
```

Wait ():

Code

```
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>

int main()
{
    printf("Before exec()\n");

    execlp("ls", "ls", NULL);

    printf("After exec()\n");

    return 0;
}
```

Result

```
ketana_sindhu@KetaZone:~$ gcc wait.c -o WAIT
ketana_sindhu@KetaZone:~$ ./WAIT
Before exec()
DisplayProcessID.c FirstFile PID.c Sindhu.c example.txt ketana sindhu
FILE.C FirstFile.c ParentAndChild.c WAIT file ketana_sindhu wait
File Fork.c ReadUser.c Wait.c file.c name wait.c
File.c HELLO.c S2Commands WeekTwo.c fork pid
FirsFile KL.c Sindhu WeekTwoONE.c hello shortcut
ketana_sindhu@KetaZone:~$ |
```

Parent And Child:

Code

```
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>

int main()
{
    pid_t pid;

    pid=fork();

    if(pid==0)
    {
        printf("I am Child Process\n");
    }
    else
    {
        printf("I am Parent Process\n");
    }

    return 0;
}
```

Result

```
ketana_sindhu@KetaZone:~$ gcc ParentAndChild.c -o parent
ketana_sindhu@KetaZone:~$ ./parent
I am Parent Process
I am Child Process
ketana_sindhu@KetaZone:~$ |
```